

Question Number	Answer	Additional guidance	Mark
3(a)	An explanation that makes reference to two of the following points: • {streamlined shape / webbed feet / wings} enable the penguin to swim (1) • {long / sharp / big / pointed} beak to {catch / eat} prey (1) • small SA: vol to reduce heat loss in cold water (1) • {layer of lipid / blubber / feathers} for {insulation / thermoregulation} (1) • colour of {feathers / skin} related to {background / habitat} (1)	ACCEPT streamlined shape to reduce {resistance / friction} ACCEPT description of webbed feet / flippers etc ACCEPT named prey IGNORE teeth ACCEPT eyes on side of head to have wider field of view ACCEPT {layer of lipid / blubber / feathers} to {withstand cold temperatures / maintain body temperature / prevent heat loss} IGNORE keep warm ACCEPT feathers for waterproofing e.g. camouflage / absorption of heat energy	(2)

Question Number	Answer	Additional guidance	Mark
3(b)(i)	An answer that makes reference to one of the following points: • when penguins from different islands breed together they produce {infertile / sterile} offspring (1) • differences in {genotype / phenotype} (1)	ACCEPT penguins from different islands cannot produce fertile offspring together ACCEPT detail of reproductive isolation e.g. they will have different courtship rituals ACCEPT molecular phylogeny ACCEPT identifying differences in molecular evidence e.g. differences in {DNA/mRNA/amino acid} (sequences), differences in protein structure IGNORE {DNA analysis / compare DNA} unqualified	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(ii)	An explanation that makes reference to five of the following points: - the different islands are located a long distance away from each other / the different penguin populations would not meet to breed / no gene flow between populations (1) {genetic variation / mutations / different alleles} in penguin population(s) (1) - different selection pressure on each island (1) {beneficial / advantageous} allele(s) may {give selective advantage in different areas / mean those individuals are more likely to survive} (1) - penguins with {beneficial / advantageous} {allele(s) / characteristics} pass these alleles onto their offspring (until the populations become genetically dissimilar to each other) (1) - the {populations / penguins} on different islands are unable to reproduce to form fertile offspring (1)	ACCEPT {geographical / allopatric} isolation (and correct descriptions of) REJECT species e.g. different {temperature / climate / predators / prey / environment} IGNORE different conditions IGNORE gene ACCEPT descriptions of advantageous characteristic linked to selection pressure ignore {survival of fittest / natural selection} unqualified IGNORE gene ACCEPT (different) populations {have different allele frequencies / different changes in allele frequency} ACCEPT {anatomical / behavioural changes} result in (different species) not being able to mate ACCEPT {reproductive / behavioural} isolation	(5)